Superpixel Spectral–Spatial Feature Fusion Graph Convolution Network for Hyperspectral Image Classification

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***Abstract*— Recently, convolutional neural networks (CNNs) have demonstrated impressive capabilities in the representation and classification of hyperspectral remote sensing images. Traditional CNNs require massive data to sufficiently train the network. To tackle this problem, graph convolutional network (GCN) has been introduced for hyperspectral image classification. GCN methods usually construct the graph from either spectral or spatial domain, which has not adequately explored the information in the joint spectral–spatial domain. In this article, we propose a superpixel spectral–spatial feature fusion graph convolution network for hyperspectral image classification (S^{3}FGCN). S^{3}FGCN can comprehensively use information in spectral, spatial, and spectral–spatial domains with limited data. Moreover, to enhance the performance, we explore a shared weights' GCN in the spectral–spatial domain. To further improve the efficiency, superpixels are used to construct the adjacency matrix. Finally, dynamic sampling is adopted to make the model focus more on difficult samples. In the experiments on four datasets, S^{3}FGCN demonstrates better accuracy compared with the state-of-the-art hyperspectral image classification methods.**

***Index Terms*— Dynamic sampling, graph convolutional network (GCN), hyperspectral image classification, spectral–spatial feature fusion.**

I. Introduction

HYPERSPECTRAL remote sensing is a multidimensional information acquisition technology that combines imaging and spectroscopy. It acquires continuous and narrow-band image data with both spatial and spectral information of the scene. As a result, hyperspectral imagery (HSI) can provide rich information about the Earth's surface with hundreds of light wavelength indexed bands. In recent decades, HSI classification has played an important role in vegetation monitoring [1], [2], mining [3], and target detection [4], [5], [6].

Most early HSI classification methods used traditional pattern recognition approaches, such as random forests [7], support vector machine (SVM) [8], Bayesian classifier [9], Gaussian maximum likelihood [10], sparse representation [11], independent component discriminant (ICA) [12], and logistic regression [13]. Since it is difficult for these methods to accurately classify HSI with only spectral features, spatial information has been used to improve the classification accuracy, e.g., using Markov random field (MRF) [14] or spectral–spatial kernel methods [15], [16]. However, these traditional pattern recognition methods are based on handcrafted features that are highly empirical and require strong domain knowledge in feature design. The deep-learning-based methods automatically learn the multilevel feature representation from the input data by building layered semantic models. These models have shown great advantages in HSI feature representation and classification [17], [18], [19], [20], [21]. Among various deep learning structures, convolutional neural networks (CNNs) have emerged as a powerful tool for HSI classification [22]. Early work adopted spectral domain reduction to enable direct application of CNNs [23], which was then extended to explore spectral–spatial HSI classification [24], [25].

Despite the success of CNNs in image classification, the existing approaches still face challenges. As the depth of the network increases, the model parameters increase quickly, and the computational complexity becomes higher. In turn, a large amount of labeled training data is required to learn a good model, but labels in HSI are difficult to obtain. The most straightforward solution to these problems is data augmentation, which generates new samples from the original samples for training, and therefore does not require additional labeled samples [26], [27]. The second solution is semisupervised learning, which learns data distribution using both labeled and unlabeled data. Among them, the semisupervised generative adversarial networks (GANs) have shown promising performance [28], [29], [30]. Transfer learning is also an effective solution. It improves the model performance using

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a priori knowledge of relevant primary task to complete the secondary task [31], [32], [33]. The last but not least solution is active learning, which uses an unlabeled sample pool in each training iteration to actively increase the size of the training data and iteratively improve the prediction performance of the classifier [34], [35].

The recent rise in graph convolutional networks (GCNs) has contributed another pathway to solving the insufficient label problem in HSI classification. By operating on a graph, training of GCN requires only a small amount of labeled data to construct the spectral or spatial relationship between nodes, and to aggregate and transform the features from the neighborhood of each graph node [36]. GCN is suitable for non-European irregular data, learning node features through hidden layers, and can better learn local features and solve the problem of missing class boundary information. Shahraki and Prasad [37] applied GCN to HSI classification by cascading GCN and 1-D neural network. Qin *et al.* [36] used GCNs in both spatial and spectral neighborhoods simultaneously and extended the original GCN to a second-order version. Yang *et al.* [38] applied a Graph SAmple and aggreGatE (GraphSAGE) network to reduce the impact of large adjacency matrices on hyperspectral classification. As the resolution of hyperspectral images becomes larger, the computational complexity increases. The superpixel segmentation methods are also commonly used in hyperspectral classification, which can effectively combine spectral and spatial features and reduce computational efforts accordingly [39], [40], [41], [42]. Jia *et al.* [39] introduced cascade superpixel regularized Gabor feature fusion for HSI classification. Yang *et al.* [42] proposed superpixel-guided discriminative low-rank representation of HSI for classification. Wan *et al.* [43] combined superpixel segmentation technology with GCN, thus reducing the computational cost.

The drawback of GCN-based methods is that the complexity of the method for constructing adjacency matrices increases rapidly in memory and time as the number of pixels in a hyperspectral image increases, making it difficult to process large images. In addition, most methods use only spectral or spatial information, or simply align and add spatial and spectral information without considering similar common information in the spectral and spatial domains [36], [37], [38], [43]. While the GraphSAGE network [38] spends a lot of time finding neighborhood relationships, this method is also unable to handle large hyperspectral images. The method of Wan *et al.* [43] uses multiscale spatial information and spectral information from spatial neighborhood to perform graph convolution. However, it does not fully use the spectral information and does not extract information common to both the spectral and spatial domains. Another approach, dual interactive graph convolutional network (DIGCN) [44], focuses on the interaction of information from two different spatial neighborhoods, but does not consider the interaction of information in the spectral dimension. To solve these problems, we propose a superpixel spectral–spatial feature fusion graph convolution network for hyperspectral image classification (S^3FGCN). We aggregate image pixels into superpixels and treat each superpixel as a node in GCN. On this basis, unlike the single feature GCN, we combine spectral features, spatial features of different scales, and spectral–spatial features to extract contextual information that is more beneficial to the final classification.

The main contributions of the proposed S^3FGCN are as follows: 1) we combine single spectral feature, spatial features of different scales, and spectral–spatial features for HSI classification; 2) we design a shared parameter matrix and the corresponding constraints for the extraction of spectral–spatial features; 3) we apply the superpixel method to reduce the computational cost, which is particularly effective for the GCN-based method; and 4) we propose dynamic sampling that pays attention to hard-to-classify sample points, leading to improved classification performance.

The rest of this article is organized as follows. Section II briefly introduces the GCN and superpixels. Section III describes the proposed S^3FGCN in detail. Section IV gives the experimental results to verify the effectiveness of the proposed method. Finally, we conclude the article and point out future work in Section V.

II. Related Work

In this section, we review the GCN model and hyperspectral superpixel segmentation. They form the basis of our proposed method.

A. GCN

GCN [45] was proposed to deal with nonstandard graph data. The feature extractor of GCN works on graph data. The extracted features can be used for graph embedding representation, node classification, link prediction, and graph classification.

Formally, an undirected graph is defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ denotes a set of vertices, $|\mathcal{V}| = N$, and $\mathcal{E}$ is a set of edges of the graph. Matrix $\boldsymbol{A} \in \mathbb{R}^{N\times N}$ indicates whether two nodes are connected or not. The feature matrix $\mathbf{X}$ is composed of feature vectors of the nodes. The GCN is also a neural network layer, and its layer-to-layer propagation model is

$$\mathbf{H}^{(l+1)} = \sigma\left(\widetilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\widetilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{H}^{(l)}\mathbf{W}^{(l)}\right) \tag{1}$$

where $\widetilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$, $\mathbf{I}$ is the identity matrix, and $\widetilde{\mathbf{D}}_{ii} = \sum_i \widetilde{\mathbf{A}}_{ij}$ is the degree matrix of $\widetilde{\mathbf{A}}$. The matrix $\mathbf{A}$ is determined by the spectral or spatial information of the HSI. $\mathbf{H}^{(l)}$ is the spectral or spatial feature representation of the l-th layer, in particular, $\mathbf{H}^{(0)} = \mathbf{X}$. $\sigma(\cdot)$ represents the activation function of the l-th layer, such as the rectified linear unit (ReLU) function or the Softmax function. $\mathbf{W}^{(l)}$ is a learnable weight matrix of the l-th layer.

B. Hyperspectral Superpixel Segmentation Method

Ren and Malik [46] proposed an image segmentation technique named superpixel segmentation. Superpixels are adjacent pixels with similar features that are successively merged into irregular pixel blocks with visual significance, where feature similarities include texture, color, brightness, and other features. Defining local regions in HSI classification is

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important for extracting spatial features from spectral–spatial models. Using superpixels as adaptive areas is more robust to noises in the data.

A commonly used superpixel segmentation method for HSI classification is simple linear iterative clustering (SLIC) [47]. The SLIC algorithm first converts the HSI from the red, green, blue (RGB) color space into the Commission Internationale de l'Eclairage (CIE)-Lab color space, and then continuously merges image pixels according to the similarity of adjacent pixels through the clustering algorithm to finally complete the segmentation of the image. Sellars *et al.* [48] proposed a superpixel method named hyper manifold SLIC (HMS), which is an extension of manifold SLIC to hyperspectral data. HMS maps an image $\mathbf{I}$ with $\mathcal{B}$ bands to a 2-D manifold $\mathcal{M} \in \mathbb{R}^{\mathcal{B}+2}$. The distance between two pixels p_x, p_y is defined using the Euclidean spectral distance and the log-Euclidean distance (LED) represented by the covariance matrix, as follows:

$$d(p_x, p_y) = \left\|\operatorname{logm}(\mathbf{C}_{p_x}) - \operatorname{logm}(\mathbf{C}_{p_y})\right\|_F + \left\|\hat{\mathbf{I}}(p_x) - \hat{\mathbf{I}}(p_y)\right\| + \frac{m}{S}\left\|p_x - p_y\right\| \quad (2)$$

where the parameter m represents the compactness of the superpixel, $\mathbf{C}_p$ is the covariance matrix, and S is used to scale the spatial distance. Assuming that an image with N pixels eventually generates K superpixels, then $S = \sqrt{N/K}$. To merge superpixel s_i with other superpixels, HMS selects the neighboring seed s_j which satisfies

$$j = \underset{s_j \in \mathcal{N}}{\operatorname{argmin}} \left\|\mathcal{P}_i^m - \mathcal{P}_j^m\right\| \quad (3)$$

where $\mathcal{P}_i^m$ and $\mathcal{P}_j^m$ are the characteristic information of the superpixels, and $s_j \in \mathcal{N}$ represents the adjacent seed set of pixel i.

III. Methodology

In this section, we introduce the S^{3}FGCN method whose framework is presented in Fig. 1. First, the HMS clustering algorithm is used to divide the hyperspectral image into many superpixels. Then these superpixels are used to obtain the hyperspectral spatial graph and the spectral graph through the K-nearest neighbors (KNN) algorithm and the neighboring relationship of the superpixels. After that, spectral and spatial convolutions are performed on different superpixels, respectively. In addition, we also extract spectral–spatial information under a shared parameter matrix and the corresponding constraints. Finally, the classifier of S^{3}FGCN generates the classification results. To focus on the training samples that are difficult to classify, we use a dynamic sampling module to select training samples.

A. Superpixel Segmentation

An HSI may have tens of thousands of pixels, so applying a GCN model at the pixel level is infeasible due to computational complexity. Superpixel segmentation can divide the image into compact superpixels, and each superpixel has strong spectral–spatial similarity. In contrast to the widely used SLIC method, the HMS method [48] combines the Euclidean spectral distance and LED, which are represented by the covariance matrix, into a clustering distance. Therefore, HMS combines the spatial and spectral data more effectively. In the superpixel merging process, similar features in the adjacent superpixels are combined to obtain a new superpixel. The clustering distance are calculated using (2). Then image segmentation uses the k-means algorithm to merge adjacent pixels according to (3) for iterative growth local clustering. Each superpixel is then fed into the GCN model as a node of the graph. In addition, the spectral characteristics of each node in the graph are obtained from the average spectral characteristics of all the pixels within each superpixel, where "all" here refers to all the pixels in training or testing. The superpixel segmentation of HSI reduces the computational complexity and retains the local information well, thus improving the accuracy of the model accordingly.

B. Spectral Graph Convolution

First, to obtain the spectral information, we construct a spectral feature graph $G_f = (\mathbf{A}_f, \mathbf{X})$, where $\mathbf{X}$ is the feature matrix obtained from HSI after superpixel segmentation and $\mathbf{A}_f$ is an adjacency matrix constructed based on the spectral similarity of the nodes of the graph $G_f = (\mathbf{A}_f, \mathbf{X})$. Specifically, constructing $\mathbf{A}_f$ requires calculating the similarity matrix $\mathbf{S} \in \mathbb{R}^{N \times N}$ of N nodes, where N is the number of superpixels. The similarity is calculated between node i and node j

$$\mathbf{S}_{ij} = e^{-\frac{\left\|\mathbf{x}_i - \mathbf{x}_j\right\|^2}{t}} \quad (4)$$

where t is a hyperparameter in the kernel function, which is set to 2 in this work. $\mathbf{x}_i$ and $\mathbf{x}_j$ are the feature vectors of superpixel i and superpixel j, respectively. For each node i, we use the KNN algorithm to select the top k similar node pairs according to the similarity $\mathbf{S}_{ij}$ between nodes i and j to construct edges in the graph $G_f = (\mathbf{A}_f, \mathbf{X})$. Finally, we get the adjacency matrix $\mathbf{A}_f$ of the spectral graph convolution network as follows:

$$\mathbf{A}_{ij} = \begin{cases} \mathbf{S}_{ij}, & \text{if } i \in \mathbf{T}_K(j) \text{ or } j \in \mathbf{T}_K(i) \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

where $\mathbf{T}_K(j)$ indicates that node i is one of the k most similar nodes to node j. Then we input the spectral feature graph $G_f = (\mathbf{A}_f, \mathbf{X})$ into GCN to get the spectral feature representation $\mathbf{H}_f^{(l)}$ of the l-th layer

$$\mathbf{H}_f^{(l+1)} = \operatorname{ReLU}\left(\widetilde{\mathbf{D}}_f^{-\frac{1}{2}} \widetilde{\mathbf{A}}_f \widetilde{\mathbf{D}}_f^{-\frac{1}{2}} \mathbf{H}_f^{(l)} \mathbf{W}_f^{(l)}\right) \quad (6)$$

where $\widetilde{\mathbf{A}}_f = \mathbf{A}_f + \mathbf{I}_f$, $\mathbf{I}_f$ is the identity matrix of the spectral feature graph, and $\widetilde{\mathbf{D}}_f$ is the degree matrix of $\widetilde{\mathbf{A}}_f$. $\mathbf{W}_f^{(l)}$ is the weight matrix of the l-th layer and $\mathbf{H}_f^{(0)} = \mathbf{X}$, initially. Using (6), we can learn further spectral information from hyperspectral images.

C. Spatial Graph Convolution

We then construct a spatial feature graph $G_s = (\mathbf{A}_s, \mathbf{X})$ based on the node feature matrix $\mathbf{X}$ to get more spatial information. First, we determine the first-order neighbor list of the

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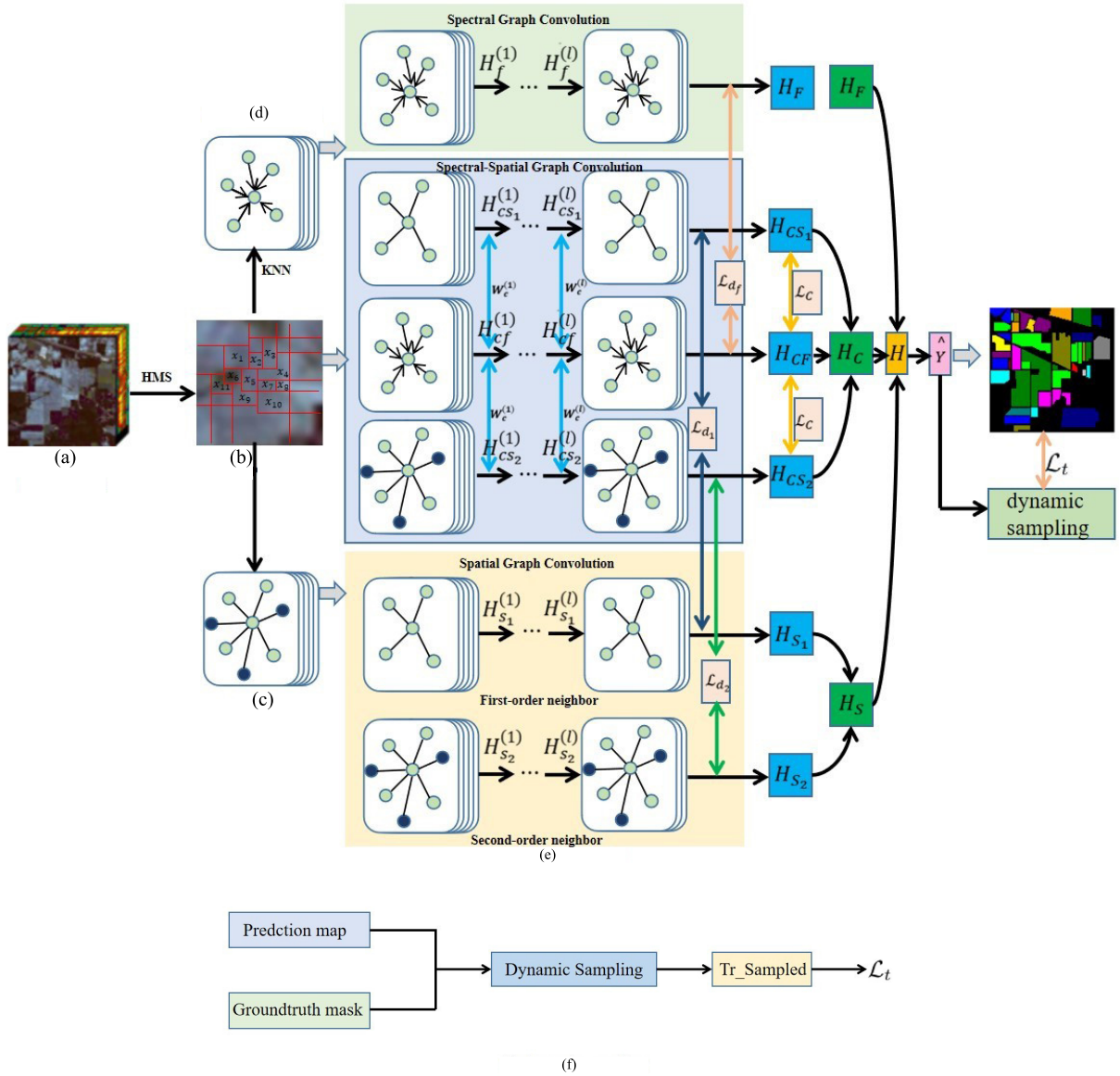


Fig. 1. Framework of the proposed S^{3}FGCN method. (a) Hyperspectral image. (b) Superpixels generated by the HMS algorithm. (c) Spatial graph formed according to the positional relationship of superpixels. (d) Spectral graph formed by the KNN algorithm using the superpixel spectral distance as a metric. (e) Three graph convolution modules. (f) Dynamic sampling module.

superpixel according to their position relationship. We obtain the coordinates $p_1(x_1, y_1)$ for the upper left endpoint and the coordinates $p_2(x_2, y_2)$ for the lower right endpoint of the superpixel region, respectively. The neighbor relationships are determined by comparing the coordinates $P_i(p_{i1}, p_{i2})$ and $P_j(p_{j1}, p_{j2})$ of two given superpixels. The adjacency matrix $\mathbf{A}_s$ can be obtained according to the corresponding neighborhood relationship

$$\mathbf{A}_{ij} = \begin{cases} c, & \text{if } P_i \in N(P_j) \text{ or } P_j \in N(P_i) \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

where c is a constant and c represents that a node is in the neighborhood of another node. The value of c can be set in different forms. One way is to use constant 1 to indicate pixel adjacency; another way is to use (4) to calculate the similarity of two pixels as in the spectral graph convolution.

Then we feed the spatial feature graph $G_s = (\mathbf{A}_s, \mathbf{X})$ into GCN to get the spatial feature representation $\mathbf{H}_s^{(l)}$ of the l-th layer

$$\mathbf{H}_s^{(l+1)} = \text{ReLU}\left(\widetilde{\mathbf{D}}_s^{-\frac{1}{2}} \widetilde{\mathbf{A}}_s \widetilde{\mathbf{D}}_s^{-\frac{1}{2}} \mathbf{H}_s^{(l)} \mathbf{W}_s^{(l)}\right) \tag{8}$$

where $\widetilde{\mathbf{A}}_s = \mathbf{A}_s + \mathbf{I}_s$, $\mathbf{I}_s$ is the identity matrix of the spatial feature graph, and $\widetilde{\mathbf{D}}_s$ is the degree matrix of $\widetilde{\mathbf{A}}_s$. ReLU is an activation function. $\mathbf{W}_s^{(l)}$ is the weight matrix of the l-th layer. The initial value of $\mathbf{H}_s^{(0)}$ is set to $\mathbf{X}$. Using (8), we can learn further spatial information about hyperspectral images.

D. Spectral–Spatial Graph Convolution

In the two separate modules described above, spatial convolution and spectral convolution are not necessarily completely unrelated. Therefore, it is not sufficient to extract

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feature from the spatial feature graph and the spectral feature graph independently. The common feature extracted from the spectral–spatial feature graph is also needed to guide the learning and fusion of features. We do so by constructing a parameter-shared GCN.

First, we use the shared parameter GCN model to extract spectral feature from graph $G_f = (\mathbf{A}_f, \mathbf{X})$

$$\mathbf{H}_{cf}^{(l+1)} = \mathrm{ReLU}\left(\widetilde{\mathbf{D}}_f^{-\frac{1}{2}}\widetilde{\mathbf{A}}_f\widetilde{\mathbf{D}}_f^{-\frac{1}{2}}\mathbf{H}_{cf}^{(l)}\mathbf{W}_c^{(l)}\right) \tag{9}$$

where $\mathbf{W}_c^{(l)}$ is the shared weight matrix of the l-th layer, $\mathbf{H}_{cf}^{(l)}$ is the common spectral embedding of nodes in the l-th layer, and $\mathbf{H}_{cf}^{(0)} = \mathbf{X}$. The spatial features from the common node in graph $G_s = (\mathbf{A}_s, \mathbf{X})$ are extracted using a shared weight matrix $\mathbf{W}_c^{(l)}$

$$\mathbf{H}_{cs_r}^{(l+1)} = \mathrm{ReLU}\left(\widetilde{\mathbf{D}}_{s_r}^{-\frac{1}{2}}\widetilde{\mathbf{A}}_{s_r}\widetilde{\mathbf{D}}_{s_r}^{-\frac{1}{2}}\mathbf{H}_{cs_r}^{(l)}\mathbf{W}_c^{(l)}\right) \tag{10}$$

where r represents the former r-th order of neighbors of $G_s = (\mathbf{A}_s, \mathbf{X})$, $\mathbf{H}_{cs_r}^{(l)}$ is the common spatial embedding of nodes in the l-th layer, and $\mathbf{H}_{cs_r}^{(0)} = \mathbf{X}$. The common embedding $\mathbf{H}_C$ of the two spaces is

$$\mathbf{H}_C = \sum_{r=1} \mathbf{H}_{CF} + \mathbf{H}_{CS_r}. \tag{11}$$

Then we concatenate $\mathbf{H}_C$, $\mathbf{H}_F$, and $\mathbf{H}_S$ obtained from different graph convolution layers and calculate the final feature $\mathbf{H}$ of the model using a linear function for classification

$$\mathbf{H} = \mathbf{W} \cdot (\mathbf{Concat}[\mathbf{H_C}, \mathbf{H_F}, \mathbf{H_S}] + \mathbf{b}). \tag{12}$$

E. Dynamic Sampling

The traditional model learning process uses all the training samples for learning. However, from a classifier perspective, there are "easy positive" training samples and "hard negative" training samples. As the model continues to learn, while the network learns a good feature representation, we expect that the misclassified samples should determine the direction of model learning. To give more attention to the hard-to-classify samples, we adopt the idea of dynamic sampling in semantic segmentation [49] to select hard-to-classify samples. During the training process, the predicted map P and the ground-truth mask G are used as input, and their intersection contains the correct samples during training. Let the intersection be $P \cap G$. For hard-to-train samples, that is, misclassified samples, subtracting the above intersection from the prediction map P leads to $P - P \cap G$. Similarly, for real graphs, misclassified samples $G - P \cap G$ can be obtained. However, the samples that are wrongly classified in the early stage of the model are not hard-to-classify samples. We need a large number of correctly classified samples to guide the model to learn features. As the network is updated and the model learns good features, fewer correct samples are needed. The training process shall involve all the hard class samples and some easy class samples to guide the learning of these classes. Therefore, we select a certain proportion of easy samples to guide the learning. The sampling is as follows:

$$\begin{aligned}\text{Tr_Sampled} &= P - P \cap G + G - P \cap G + \text{ratio} \cdot P \cap G \\ &= P \cup G - (1 - \text{ratio}) \cdot P \cap G\end{aligned} \tag{13}$$

where Tr_Sampled represents the samples used to train the cross-entropy loss. For the above sampling, instead of taking a fixed value, the ratio between easy samples should be reduced as the model training deepens. We take the inverse proportional function below for dynamic sampling

$$\text{ratio} = 1 - (\text{iter}/\text{Epoch}) \tag{14}$$

where iter is the current training epoch, and Epoch is the number of training epoch.

F. Objective Function

Through the above model framework, we can learn the common features from the spectral, spatial, and spectral–spatial domains. But we also need to add constraints to ensure the effectiveness of the features and that the spectral graph convolution, spatial graph convolution, and spectral-spatial shared graph convolution can learn different embedded information. Our proposed method uses two constraints [50] to ensure that the information extracted is beneficial to the final classification.

1) Similarity Constraint: The outputs $\mathbf{H}_{CF}$ and $\mathbf{H}_{CS}$ obtained by the common graph convolution module are expected to have certain level of similarity in the extracted features. The similarity is measured by the distance between $\mathbf{H}_{CF}$ and $\mathbf{H}_{CS}$. First, we normalize the two outputs $\mathbf{H}_{CF}$ and $\mathbf{H}_{CS}$ to get the normalized outputs $\mathbf{H}_{CFnor}$ and $\mathbf{H}_{CSnor}$. The similarity matrix $\mathbf{S}_F$ and $\mathbf{S}_S$ between each pair of nodes can be obtained through the normalized matrix

$$\begin{aligned}\mathbf{S}_F &= \mathbf{H}_{CF\mathrm{nor}} \cdot \mathbf{H}_{CF\mathrm{nor}}^T \\ \mathbf{S}_S &= \mathbf{H}_{CS\mathrm{nor}} \cdot \mathbf{H}_{CS\mathrm{nor}}^T.\end{aligned} \tag{15}$$

The output obtained through the shared GCN should be similar, which creates a similarity constraint

$$\mathcal{L}_{\mathrm{sim}} = \|\mathbf{S}_S - \mathbf{S}_F\|_F^2. \tag{16}$$

2) Disparity Constraint: We expect to extract two different features that are beneficial to the final classification from $\mathbf{H}_{S_r}$ obtained by the spatial convolution and $\mathbf{H}_{CS_r}$ obtained by the common convolution. To this end, we adopt the Hilbert–Schmidt independence criterion (HSIC) [51] as in adaptive multi-channel graph convolutional networks (AM-GCN) [50]. HSIC is a measure of the independence between two variables to enhance the difference between them. The constraint on $\mathbf{H}_{S_r}$ and $\mathbf{H}_{CS_r}$ is

$$\mathrm{HSIC}\left(\mathbf{H}_{S_r}, \mathbf{H}_{CS_r}\right) = (n-1)^{-2}\,\mathrm{tr}\left(\mathbf{K}_{S_r}\mathbf{J}\mathbf{K}_{CS_r}\mathbf{J}\right) \tag{17}$$

where $\mathbf{K}_{S_r} \in \mathbb{R}^{n\times n}$ and $\mathbf{K}_{CS_r} \in \mathbb{R}^{n\times n}$ have entries $k_{S_r,m\ n} = k_{S_r}(\mathbf{H}_{C_m}, \mathbf{H}_{C_n})$ and $k_{CS_r,m\ n} = k_{CS_r}(\mathbf{H}_{CS_{rm}}, \mathbf{H}_{CS_{rn}})$. $\mathbf{K}$ is the Gram matrices, and k is the kernel functions. $\mathbf{J} = \mathbf{I}_n - (1/n)\mathbf{1}_n\mathbf{1}_n^T$ is the centering matrix and $\mathbf{1}_n$ is a column vector.

Similarly, we expect to extract two different features from $\mathbf{H}_F$ obtained by the spectral convolution and $\mathbf{H}_{CF}$ obtained by the common convolution. Their constrained $\mathrm{HSIC}(\mathbf{H}_F, \mathbf{H}_{CF})$ is

$$\mathrm{HSIC}(\mathbf{H}_F, \mathbf{H}_{CF}) = (n-1)^{-2}\,\mathrm{tr}(\mathbf{K}_F\mathbf{J}\mathbf{K}_{CF}\mathbf{J}). \tag{18}$$

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Then the total loss of disparity constraint $\mathcal{L}_{\text{dis}}$ is the sum of $\text{HSIC}(\mathbf{H}_F, \mathbf{H}_{CF})$ and $\text{HSIC}(\mathbf{H}_{S_r}, \mathbf{H}_{CS_r})$

$$\mathcal{L}_{\text{dis}} = \sum_{r=1} \text{HSIC}\left(\mathbf{H}_{S_r}, \mathbf{H}_{CS_r}\right) + \text{HSIC}(\mathbf{H}_F, \mathbf{H}_{CF}). \quad (19)$$

3) Objective Function: Our final task is to classify and predict n nodes. The predicted value by the network for each node is $\hat{\mathbf{Y}} = [\hat{y}_{ic}] \in \mathbb{R}^{n \times C}$, and then $\hat{\mathbf{Y}}$ can be expressed as

$$\hat{\mathbf{Y}} = \text{softmax}(\mathbf{H}) \quad (20)$$

where $\mathbf{H}$ is the embedding obtained by (12). For the training set, we obtain Tr_Sample by dynamic sampling. For each node l used for training, its true label is $\mathbf{Y}_{l_{\text{Tr}}}$ and the predicted label is $\hat{\mathbf{Y}}_{l_{\text{Tr}}}$. Then we compute the cross-entropy loss on the dynamically sampled nodes in the training set

$$\mathcal{L}_{\text{Tr_Sample}} = -\sum_{l \in L} \sum_{j=1}^{C} Y_{l_{\text{Tr}}^j} \ln \hat{Y}_{l_{\text{Tr}}^j} \quad (21)$$

where C is the number of categories of nodes.

Finally, we combine the cross-entropy loss, consistency loss and disparity loss to get the final loss function

$$\mathcal{L} = \mathcal{L}_{\text{Tr_Sample}} + \alpha \mathcal{L}_{\text{sim}} + \beta \mathcal{L}_{\text{dis}} \quad (22)$$

where α and β are the weights for balancing the loss. Algorithm 1 summarizes the proposed S^{3}FGCN classification method.

Algorithm 1 S^{3}FGCN

Input: Input hyperspectral image; learning rate η; number of iterations EPOCHS; number of superpixels K; number of graph convolutional layers L; the former r-th order of neighbors of $G_s = (\mathbf{A}_s, \mathbf{X})$; loss function weights (α, β) in (22);

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Segment HSI into K superpixels using HMS superpixel segmentation algorithm;
Construct the initial adjacency matrices of the spectral and spatial domains according to (7) and (4);
Train the S^3FGCN model;
for epoch = 1 to EPOCHS do
    Calculate the spectral graph convolution of the l-th layer H_F^(l) according to (6);
    Calculate the spatial graph convolution of the l-th layer H_Sk^(l) according to (8);
    Calculate the spectral–spatial graph convolution of the l-th layer H_C^(l) according to (11);
    Combine the above feature to get the l-th layer H^(l) according to (12);
    Calculate two constraints L_sim, L_dis and sample training set nodes according to (13);
    Calculate the loss between the predicted value and the true label according to (21);
    Add the above losses to get the final loss L according to (22);
    Update the weight matrices;
end for
```

Output: Predicted label for each pixel.

IV. Experimental Results

In this section, we report the performance of the proposed S^{3}FGCN method on four datasets, i.e., Indian Pines, Pavia University, Salinas, and the University of Houston. Three performance metrics have been used: overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa).

A. Hyperspectral Datasets

1) Indian Pines Dataset: The Indian Pines dataset was captured in Indiana, USA, in 1992 using an AVIRIS sensor. The sensor captured images in 400–2500 nm with 220 consecutive bands, and the image size was cropped to 145 × 145 pixels. Bands 104–108, 150–163, and 220 were removed due to water absorption, leaving 200 bands for the experiments. There are 16 categories in the dataset, whose numbers of training and testing pixels are listed in Table I, with labels shown in Fig. 2.

2) Pavia University Dataset: The Pavia University dataset was captured over Pavia, Italy, in 2003. The captured image has 115 bands from 430 to 860 nm at a spatial resolution of 1.3 m. After 12 bands were removed due to heavy noise, 103 bands were used for experiments. The image size is 610 × 340, containing mostly background pixels. The foreground has 42 776 pixels in nine categories, as summarized in Table II. Fig. 3 shows the labeled samples.

3) Salinas Hyperspectral Dataset: The Salinas dataset was captured by the AVIRIS sensor at the Salinas Valley, USA. Its spatial resolution is 3.7 m. The experiments used 204 bands after removing 20 water absorption bands. The size of this image is 512 × 217. Table III lists the number of training and testing pixels of each class, and Fig. 4 shows the labeled samples.

TABLE I
Summary of Training and Testing Samples in the Indian Pines Dataset

Class No.	Class Name	Training	Testing
1	Alfalfa	30	31
2	Corn-notill	30	1398
3	Corn-mintill	30	800
4	Corn	30	207
5	Grass-pasture	30	453
6	Grass-trees	30	700
7	Grass-pasture-mowed	15	13
8	Hay-windrowed	30	448
9	Oats	15	5
10	Soybean-notill	30	942
11	Soybean-mintill	30	2425
12	Soybean-clean	30	563
13	Wheat	30	175
14	Woods	30	1235
15	Buildings-Grass-Trees-Drives	30	356
16	Stone-Steel-Towers	30	63
-	Total	440	9809

4) University of Houston Dataset: The University of Houston dataset was captured over the University of Houston University and its neighboring areas. The size of the image is 349 × 1905 pixels, with a spatial resolution of 2.5 m. The image has 144 spectral bands from 380 to 1050 nm. There are 15 categories in the image as displayed in Fig. 5. Table IV shows the numbers of training and testing pixels of each class.

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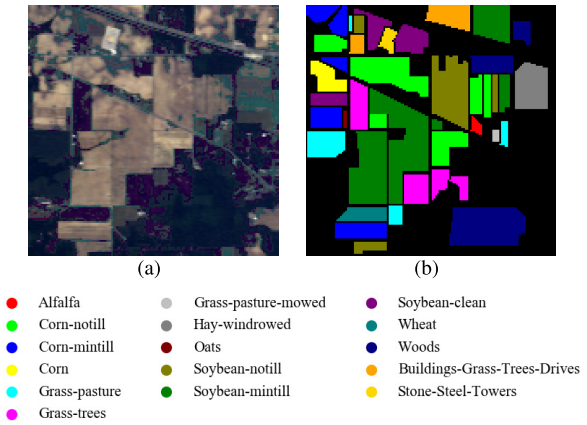


Fig. 2. Indian Pines dataset. (a) False color image. (b) Ground-truth map.

TABLE II

SUMMARY OF TRAINING AND TESTING SAMPLES IN THE PAVIA UNIVERSITY DATASET

Class No.	Class Name	Training	Testing
1	Asphalt	30	6601
2	Meadows	30	18619
3	Gravel	30	2069
4	Trees	30	3034
5	Painted metal sheets	30	1315
6	Bare Soil	30	4999
7	Bitumen	30	1300
8	Self-Blocking Bricks	30	3652
9	Shadows	30	917
-	Total	270	42506

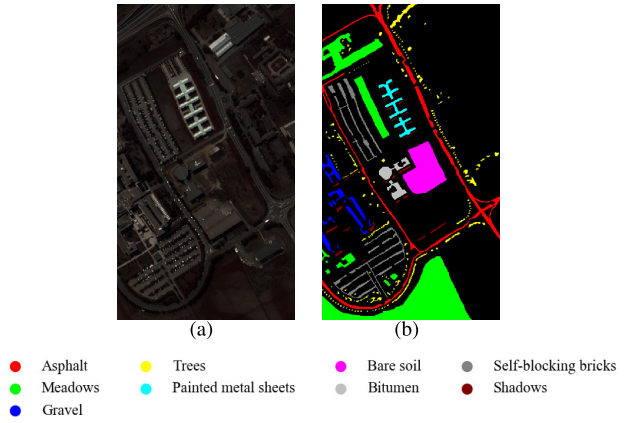


Fig. 3. Pavia University dataset. (a) False color image. (b) Ground-truth map.

B. Parameter Settings

The proposed algorithm was implemented via PyTorch with an Adam optimizer. In all, 30 pixels of each class were randomly selected for training in each dataset, and 15 pixels were selected if a class has less than 30 pixels. The remaining pixels of each type were all used for testing. This allows the extraction of multiscale information. We set the initial learning rate to 0.001 and the total number of training epochs to 5000.

TABLE III

SUMMARY OF TRAINING AND TESTING SAMPLES IN THE SALINAS HYPERSPECTRAL DATASET

Class No.	Class Name	Training	Testing
1	Brocoli_green_weeds_1	30	1979
2	Brocoli_green_weeds_2	30	3696
3	Fallow	30	1946
4	Fallow_rough_plow	30	1364
5	Fallow_smooth	30	2648
6	Stubble	30	3929
7	Celery	30	3549
8	Grapes_untrained	30	11241
9	Soil_vinyard_develop	30	6173
10	Corn_senesced_green_weeds	30	3248
11	Lettuce_romaine_4wk	30	1038
12	Lettuce_romaine_5wk	30	1897
13	Lettuce_romaine_6wk	30	886
14	Lettuce_romaine_7wk	30	1040
15	Vinyard_untrained	30	7238
16	Vinyard_vertical_trellis	30	1777
-	TOTAL	480	53649

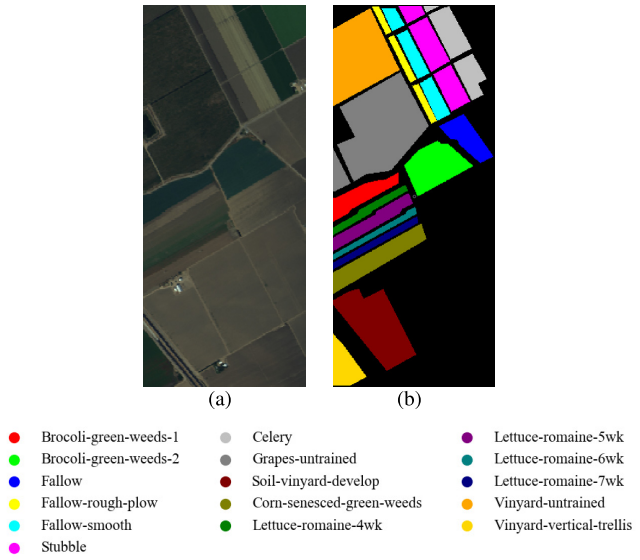


Fig. 4. Salinas dataset. (a) False color image. (b) Ground-truth map.

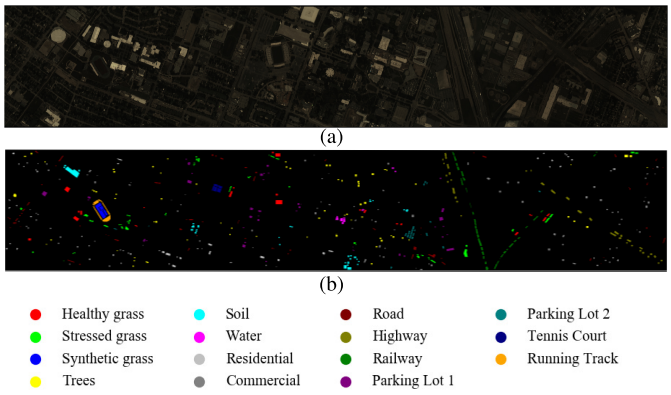


Fig. 5. University of Houston dataset. (a) False color image. (b) Ground-truth map.

Then we used a dynamic decay strategy to decay the learning rate with the number of training epochs.

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TABLE IV
SUMMARY OF TRAINING AND TESTING SAMPLES IN THE UNIVERSITY OF HOUSTON DATASET

Class No.	Class Name	Training	Testing
1	Healthy grass	30	1221
2	Stressed grass	30	1224
3	Synthetic grass	30	667
4	Trees	30	1214
5	Soil	30	1212
6	Water	30	295
7	Residential	30	1238
8	Commercial	30	1214
9	Road	30	1222
10	Highway	30	1197
11	Railway	30	1205
12	Parking Lot 1	30	1203
13	Parking Lot 2	30	439
14	Tennis Court	30	398
15	Running Track	30	630
-	TOTAL	450	14579

C. Impact of Hyperparameters

Since our method used HMS, to illustrate the effectiveness of the HMS method, we show the superpixel segmentation of each dataset in Fig. 6. From Fig. 6, we can see that the superpixels are reasonable. In our proposed method, r refers to the highest order of neighbors adopted for feature learning. To find the best r, we conducted experiments on four datasets and set the r from 1 to 5. The results are shown in Fig. 7. It can be seen that the model works best when $r = 2$, which means the best performance is obtained for the first-order and second-order neighborhoods. When r becomes larger, the learned information may be redundant, which is not conducive to feature learning, resulting in a decrease in accuracy.

c is another parameter that represents that a node is in the neighborhood of another node. The value of c can be set in different methods. One method is to use a constant 1 to indicate that the pixels are adjacent, and another method is to use (4) to calculate the similarity of two pixels. We used these two methods to calculate the accuracies displayed in Table V. According to the experimental results, we set $c = 1$. In addition, we studied the impact of the parameters on the loss function in (22). These included α which is the weight coefficient of consistency constraint and β which is the weight coefficient of disparity constraint. We also calculated the OA on each dataset by changing the parameters (α and β) from $1e-8$ to $1e-1$. As shown in Table VI, we selected the top five OA rankings in each dataset for labeling. It can be seen that the average performance of the model in the first three datasets is the best when the parameters (α and β) are set to ($1e-3$, $1e-7$). Because the foreground data of Houston dataset are sparser than other datasets, their selected parameters for (α and β) were set to ($1e-5$ and $1e-6$), respectively.

D. Classification Results

We compared our method with a few traditional machine learning methods, deep learning methods, and GCN-based methods in hyperspectral image classification. These include multilayer perception (MLP) and SVM based on radial basis function (RBF) kernel [52], which are the traditional machine learning methods. In the deep-learning-based methods, we compared two CNN-based methods, including diverse region-based deep CNN (DRCNN) [20] and 3D CNN-based feature extraction method (3DCNN) [25]. Then we compared the three models GCN [45], S^{2}GCN [36], and S^{2}GraphSAGE [38] directly based on the GCN framework. Finally, we added comparison with multiscale dynamic graph convolutional network (MDGCN) [43] and DIGCN [44], which were used in conjunction with GCN and superpixels. We trained each model on the four datasets for ten times with random data splitting and report the mean testing results.

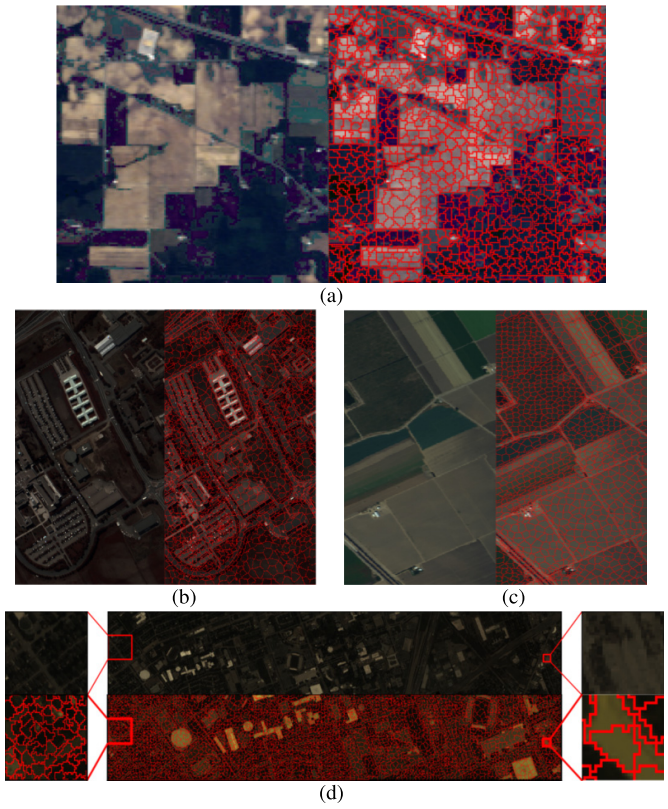


Fig. 6. Superpixel settings for different datasets. (a) Indian Pines dataset ($K = 1200$). (b) Pavia University dataset ($K = 1500$). (c) Salinas dataset ($K = 2400$). (d) University of Houston dataset ($K = 15\,000$).

1) Results on the Indian Pines Dataset: Table VII lists the test results of our proposed method and different methods on the Indian Pines dataset, among which the best performance is shown in bold. The table shows that our S^{3}FGCN method shows great advantages in various metrics and their standard deviations are relatively small. Among them, the accuracy of OA is 1.81% higher than the DIGCN model, which fully shows that the fusion of first- or second-order spatial neighborhoods, multiembeddings, and the guidance of dynamic sampling improve the expressive ability of information. Fig. 8 shows the visualization results of our proposed method S^{3}FGCN and other methods. It can be seen from the figure that the GCN-based method handles the boundary information better. Compared with MDGCN and DIGCN, our S^{3}FGCN method has better classification results and smoother visual effect.

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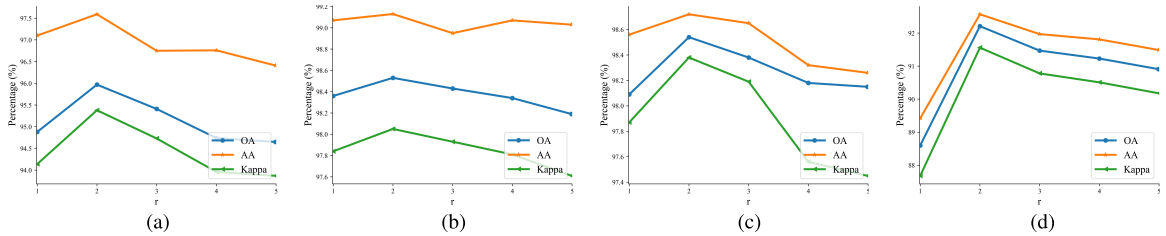


Fig. 7. OA (%), AA (%), and Kappa coefficient (%) achieved by different r. (a) Indian Pines dataset. (b) Pavia University dataset. (c) Salinas dataset. (d) University of Houston dataset.

TABLE V
OA (%), AA (%), AND KAPPA COEFFICIENT (%) ACHIEVED BY SETTING OF PARAMETER c IN SPATIAL GRAPH CONVOLUTION

$C=1$	IP	PaviaU	Salinas	Houston
OA	**95.97±0.63**	**98.53±0.84**	**98.54±0.16**	**92.21±0.46**
AA	**97.59±0.41**	**99.13±0.31**	**98.72±0.27**	**92.57±0.57**
Kappa	**95.38±0.72**	**98.05±1.09**	**98.38±0.18**	**91.56±0.50**
$C=e^{-\frac{\|x_i-x_j\|^2}{t}}$	IP	PaviaU	Salinas	Houston
OA	95.56±0.64	97.63±0.86	97.88±0.39	90.17±0.98
AA	97.34±0.33	98.20±0.51	98.40±0.35	90.99±0.92
Kappa	94.92±0.73	96.86±1.13	97.64±0.43	89.37±1.06

2) Results on Pavia University Dataset: Table VIII lists the test results of our proposed method and different methods on the Pavia University dataset, where the best results are shown in bold. Our method has greatly improved performance over the alternative methods. We achieved 2.44% higher OA than the MDGCN method based on superpixel segmentation and GCN, and 5.29% higher OA than the DIGCN method. Our method fuses information at different scales and their embeddings to better learn hard-to-classify sample points through dynamic sampling. Compared with other methods, our S^{3}FGCN method has better classification results and smoother visual effect as shown in Fig. 9.

3) Results on Salinas Dataset: Table IX lists the test results of our proposed method and different methods on the Salinas dataset. Compared with the second best-performing DIGCN method, our method outperforms it in OA, AA, and Kappa. DIGCN does not perform well on several classes which are greatly improved by our method, such as Fallow-smooth, Lettuce-romaine-5wk, and Lettuce-romaine-6wk. Our method achieved 0.93% higher OA than the DIGCN model, indicating that our method is better in the final information extraction for classification. Fig. 10 shows the visualization results of the S^{3}FGCN method and other methods. It can be seen from the figure that the GCN-based method handles the boundary information better.

4) Results on the University of Houston Dataset: Table X shows the experimental results on the University of Houston dataset. The classification performance of the deep-learning-based method is much better than that of the traditional machine learning method. In the MDGCN and DIGCN models that introduce superpixels, local information is better fused through superpixels and the complexity is effectively reduced. Their accuracies are also better than other methods. Our method not only has the above advantages but also better extracts common features. Fig. 11 demonstrates a comparison of the visualized classification results and ground-truth maps of our proposed method and other methods. It is clear that our S^{3}FGCN method produced better results in terms of accuracy and smoothness of the boundary.

The above experimental results show that our method achieves the best performance on all the four datasets. thanks to its stronger capability to extract spectral and spatial information. Our method extracts the features separately from the spectral and spatial domains, extracts the common features in the joint spectral–spatial domain, and finally performs feature aggregation, leading to a stronger information expression ability. Moreover, our method not only uses the spectral–spatial features of different scales but also uses the shared matrix to extract the same information of the spectral and spatial of different scales, which is superior to other methods. In addition, we pay more attention to the sample points that are difficult to classify through the process of dynamic sampling, which is beneficial to the network model in learning information at different scales.

E. Impact of the Number of Labeled Examples

To verify that our method works well with different sample sizes, we compared all the methods with the per-class training samples changing from 5 to 30 with a step of 5. Ten experiments were performed on each method, and the average OA was recorded. Fig. 12 shows the performance of all the methods with different sample sizes. It can be seen from the figure that as the number of samples increases, the accuracy of the experiment also increases. In the traditional methods and deep learning methods, the smaller the number of samples, the worse the performance, because they all require a large number of labeled samples for learning, and fewer samples are easy to fall into local optimum. However, the performance of the GCN-based methods decreases less as the number of samples decreases, because the GCN methods do not require label guidance when establishing graph relationships, but can also learn good information with few samples. In all the models, our method performs the best under all the sample sizes, which again proves that our method is effective in hyperspectral image classification due to its strong spectral and spatial information extraction capability.

F. Ablation Study

Our model includes three graph convolution modules: spectral graph convolution (spectral-GCN), spatial graph convolution (spatial-GCN), and spectral–spatial graph convolution

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TABLE VI

IMPACT OF THE WEIGHTS OF LOSS FUNCTION

(α, β)	Indian	Pavia	Salinas	Houston	(α, β)	Indian	Pavia	Salinas	Houston	(α, β)	Indian	Pavia	Salinas	Houston
$(1e-8, 1e-8)$	95.12	98.59	**98.75**	92.00	$(1e-8, 1e-7)$	96.01	98.27	98.45	91.92	$(1e-8, 1e-6)$	95.85	98.70	98.59	91.88
$(1e-8, 1e-5)$	95.44	97.94	98.36	91.78	$(1e-8, 1e-4)$	94.55	96.89	98.29	**92.12**	$(1e-8, 1e-3)$	93.94	89.82	97.55	91.19
$(1e-8, 1e-2)$	92.73	83.63	95.59	89.48	$(1e-8, 1e-1)$	85.84	79.74	89.21	88.33					
$(1e-7, 1e-8)$	95.39	98.60	98.60	91.72	$(1e-7, 1e-7)$	**96.39**	98.30	98.39	91.52	$(1e-7, 1e-6)$	95.75	98.64	98.47	91.66
$(1e-7, 1e-5)$	95.47	96.81	98.48	91.52	$(1e-7, 1e-4)$	94.77	96.24	98.47	92.02	$(1e-7, 1e-3)$	93.79	90.87	97.82	91.03
$(1e-7, 1e-2)$	91.95	84.23	93.21	89.07	$(1e-7, 1e-1)$	86.09	80.42	90.32	87.62					
$(1e-6, 1e-8)$	95.34	98.65	98.58	91.95	$(1e-6, 1e-7)$	96.00	98.24	98.43	92.10	$(1e-6, 1e-6)$	95.60	98.68	98.40	91.95
$(1e-6, 1e-5)$	95.14	97.29	98.64	92.04	$(1e-6, 1e-4)$	94.29	96.77	98.11	91.77	$(1e-6, 1e-3)$	94.30	88.52	97.27	91.37
$(1e-6, 1e-2)$	92.41	83.85	93.31	88.53	$(1e-6, 1e-1)$	85.91	79.76	91.79	86.14					
$(1e-5, 1e-8)$	95.61	98.21	98.45	92.15	$(1e-5, 1e-7)$	**96.14**	98.65	98.45	91.99	$(1e-5, 1e-6)$	95.31	**98.74**	98.54	**92.25**
$(1e-5, 1e-5)$	95.43	98.05	98.68	91.90	$(1e-5, 1e-4)$	94.99	96.47	98.29	91.85	$(1e-5, 1e-3)$	93.49	88.27	97.73	90.66
$(1e-5, 1e-2)$	92.72	84.93	92.88	88.31	$(1e-5, 1e-1)$	86.22	79.59	91.34	86.92					
$(1e-4, 1e-8)$	95.52	98.55	98.47	91.78	$(1e-4, 1e-7)$	95.91	98.31	**98.80**	**92.18**	$(1e-4, 1e-6)$	95.89	**98.77**	98.33	91.92
$(1e-4, 1e-5)$	95.47	97.99	98.70	91.84	$(1e-4, 1e-4)$	94.59	97.25	98.26	91.84	$(1e-4, 1e-3)$	93.88	91.07	97.87	91.55
$(1e-4, 1e-2)$	91.94	83.56	93.50	89.46	$(1e-4, 1e-1)$	85.68	80.23	91.33	87.48					
$(1e-3, 1e-8)$	95.00	97.30	98.49	91.99	$(1e-3, 1e-7)$	**96.43**	**98.73**	**98.81**	91.68	$(1e-3, 1e-6)$	96.01	98.70	98.60	92.00
$(1e-3, 1e-5)$	95.38	98.00	98.62	92.06	$(1e-3, 1e-4)$	94.70	96.79	98.22	91.87	$(1e-3, 1e-3)$	94.68	91.02	97.55	90.88
$(1e-3, 1e-2)$	92.31	83.85	94.31	88.87	$(1e-3, 1e-1)$	85.49	79.93	90.18	87.27					
$(1e-2, 1e-8)$	95.81	98.38	**98.72**	91.73	$(1e-2, 1e-7)$	96.10	**98.72**	98.61	91.78	$(1e-2, 1e-6)$	95.90	**98.71**	98.66	92.00
$(1e-2, 1e-5)$	95.46	97.38	98.50	91.90	$(1e-2, 1e-4)$	94.63	96.25	97.86	**92.11**	$(1e-2, 1e-3)$	93.49	89.29	97.79	90.73
$(1e-2, 1e-2)$	92.76	83.86	93.57	89.72	$(1e-2, 1e-1)$	84.49	80.00	89.42	87.88					
$(1e-1, 1e-8)$	**96.27**	98.70	98.25	91.98	$(1e-1, 1e-7)$	**96.35**	98.61	**98.79**	90.54	$(1e-1, 1e-6)$	95.78	98.63	98.60	**92.14**
$(1e-1, 1e-5)$	95.18	98.01	98.68	92.08	$(1e-1, 1e-4)$	94.91	96.11	98.17	91.88	$(1e-1, 1e-3)$	93.74	90.62	97.40	90.12
$(1e-1, 1e-2)$	93.26	83.10	93.90	88.29	$(1e-1, 1e-1)$	85.44	79.60	91.44	87.25					

TABLE VII

RESULTS OF DIFFERENT METHODS ON THE INDIAN PINES DATASET

ID	MLP	SVM (RBF)	3DCNN	DRCNN	GCN	S^2GCN	S^2GraphSAGE	MDGCN	DIGCN	S^3FGCN
1	30.43±1.78	30.87±4.71	**100.00±0.00**	**100.00±0.00**	96.97±5.25	99.15±1.90	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
2	39.99±10.78	27.65±3.36	73.61±7.36	81.73±6.17	43.89±4.36	67.99±5.76	74.21±0.92	85.68±3.51	90.92±4.27	**92.25±1.94**
3	51.93±13.28	36.05±12.09	72.44±8.37	86.66±3.65	50.45±8.66	86.11±7.47	88.28±4.15	84.08±2.85	94.87±3.92	**96.19±2.51**
4	60.20±6.97	71.85±5.43	94.79±4.59	98.95±1.12	**99.31±0.60**	96.88±3.17	96.18±3.05	98.84±2.14	98.32±1.93	99.13±1.61
5	44.45±19.89	50.35±20.97	92.23±0.74	94.97±2.64	69.18±10.66	93.23±3.03	94.45±3.31	93.51±4.14	95.29±3.44	**97.07±2.22**
6	69.63±7.83	79.37±3.56	**99.36±0.36**	99.12±0.43	90.03±2.07	98.93±0.81	98.39±2.40	97.23±2.10	95.95±1.85	97.16±1.26
7	39.29±4.61	42.86±2.52	**100.00±0.00**	**100.00±0.00**	97.44±4.44	97.42±4.44	**100.00±0.00**	89.5±9.26	96.36±4.03	**100.00±0.00**
8	82.01±3.08	76.74±1.68	**100.00±0.00**	99.92±0.11	93.07±0.80	99.46±0.13	99.69±0.13	99.82±0.38	99.84±0.37	**100.00±0.00**
9	13.33±6.29	10.00±4.47	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
10	64.27±6.02	48.58±11.15	76.34±13.53	89.82±2.88	78.68±1.62	86.41±7.89	90.18±5.37	90.15±4.49	89.58±4.95	**92.88±2.68**
11	35.49±1.87	51.57±16.71	65.67±9.34	64.45±8.74	62.33±4.39	76.31±3.35	80.32±4.71	93.39±2.99	91.75±3.52	**95.38±1.42**
12	23.10±12.32	16.26±7.00	82.11±5.02	82.72±4.93	49.69±5.59	88.20±3.51	93.49±1.87	92.57±2.56	**93.81±3.73**	92.95±3.38
13	82.60±1.08	80.98±1.25	**100.00±0.00**	**100.00±0.00**	98.75±0.00	97.29±0.72	98.12±0.63	99.43±1.21	99.65±0.68	98.52±2.00
14	71.09±11.77	73.68±16.72	93.19±3.28	91.74±2.25	74.86±6.23	94.04±0.70	94.35±0.94	97.39±3.47	98.94±1.73	**100.00±0.00**
15	39.72±4.37	22.07±7.04	92.86±6.24	98.53±1.67	83.09±7.21	92.04±5.23	88.07±9.37	99.02±0.60	98.58±1.60	**100.00±0.00**
16	63.08±2.24	63.66±3.83	**100.00±0.00**	**100.00±0.00**	97.92±3.61	99.21±1.37	98.61±2.41	97.14±2.09	99.48±0.72	**100.00±0.00**
OA(%)	50.75±0.95	50.32±2.73	80.39±0.76	83.91±1.87	67.20±1.37	84.76±0.69	87.43±1.48	93.56±0.91	94.16±1.07	**95.97±0.63**
AA(%)	50.66±1.56	48.70±1.67	90.16±1.55	93.01±0.52	80.35±1.60	92.11±1.09	93.40±0.92	95.36±0.50	96.41±0.54	**97.59±0.41**
Kappa(%)	45.58±1.19	44.20±2.43	77.69±0.93	81.77±2.06	62.72±1.48	82.58±0.7	85.66±1.66	92.64±1.03	93.34±1.21	**95.38±0.72**

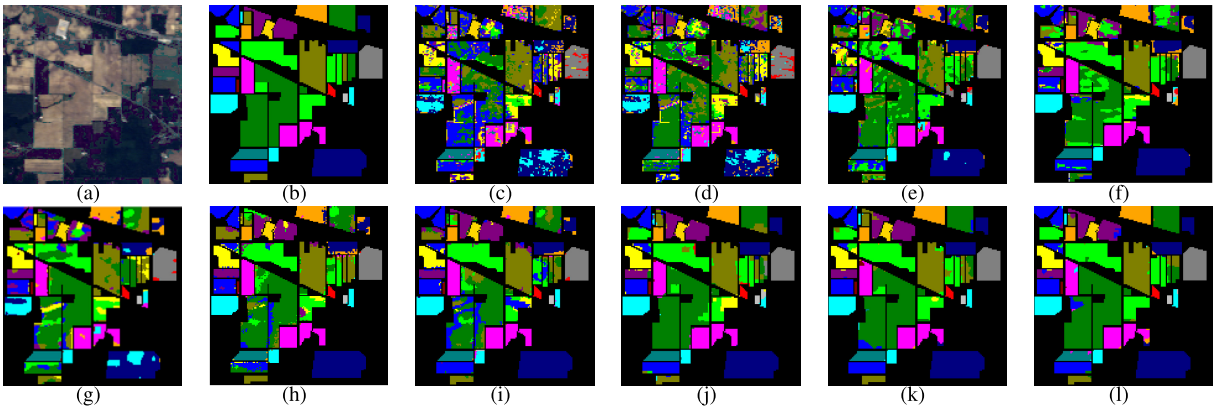


Fig. 8. Classification maps obtained by different methods on the Indian Pines dataset. (a) False color image. (b) Ground-truth. (c) MLP. (d) SVM (RBF). (e) 3DCNN. (f) DRCNN. (g) GCN. (h) S^2GCN. (i) S^2GraphSAGE. (j) MDGCN. (k) DIGCN. (l) S^3FGCN.

TABLE VIII

RESULTS OF DIFFERENT METHODS ON THE PAVIA UNIVERSITY DATASET

ID	MLP	SVM (RBF)	3DCNN	DRCNN	GCN	S^2GCN	S^2GraphSAGE	MDGCN	DIGCN	S^3FGCN
1	64.37±3.05	59.56±2.82	77.86±3.92	90.99±2.06	69.78±4.71	92.45±3.58	84.87±4.29	97.10±1.88	84.59±3.64	**98.82±1.11**
2	54.66±1.17	55.15±7.53	83.40±5.31	96.71±1.92	54.10±10.54	87.32±4.12	95.29±1.06	94.21±2.53	94.75±2.79	**97.77±1.91**
3	47.55±1.71	34.81±9.84	74.56±5.17	84.56±4.65	69.69±4.48	87.98±4.53	85.96±1.83	98.50±1.79	95.10±3.31	**99.29±0.65**
4	92.19±2.25	93.68±2.24	92.01±1.15	97.01±0.56	91.23±7.03	91.25±0.58	83.63±0.98	96.89±1.28	84.89±4.02	**97.62±1.85**
5	98.83±0.03	97.01±0.22	**100.00±0.00**	99.97±0.03	98.74±0.11	**100.00±0.00**	99.52±0.40	98.05±1.46	99.17±0.61	**100.00±0.00**
6	62.66±4.82	69.91±9.76	80.02±3.01	92.11±6.18	65.34±10.53	88.34±5.51	85.85±6.24	97.83±2.14	**99.97±0.05**	99.85±0.35
7	90.02±2.21	90.45±1.43	95.27±3.06	94.37±1.77	86.64±4.68	99.01±0.93	96.78±1.40	99.71±0.05	98.23±1.77	**99.99±0.02**
8	66.38±0.51	77.64±8.73	91.48±3.54	93.68±3.86	72.26±2.63	89.53±3.31	94.16±0.49	**98.94±0.97**	93.94±2.67	97.25±2.92
9	99.74±0.10	96.73±0.12	**100.00±0.00**	97.92±0.18	99.93±0.06	98.75±0.68	97.67±0.19	96.81±1.11	93.21±5.05	99.89±0.00
OA(%)	63.87±0.17	64.60±2.37	84.24±2.65	94.87±1.85	66.19±3.43	90.10±0.95	91.41±0.37	96.09±1.32	93.24±1.24	**98.53±0.84**
AA(%)	75.16±0.01	74.99±0.33	88.29±0.49	93.82±0.28	78.63±1.23	92.34±0.56	91.53±0.72	97.38±0.53	93.76±0.90	**99.13±0.31**
Kappa(%)	55.56±0.42	56.61±2.17	79.63±3.19	93.16±2.15	58.39±3.28	87.21±2.15	88.61±0.53	94.87±1.71	91.14±1.58	**98.05±1.09**

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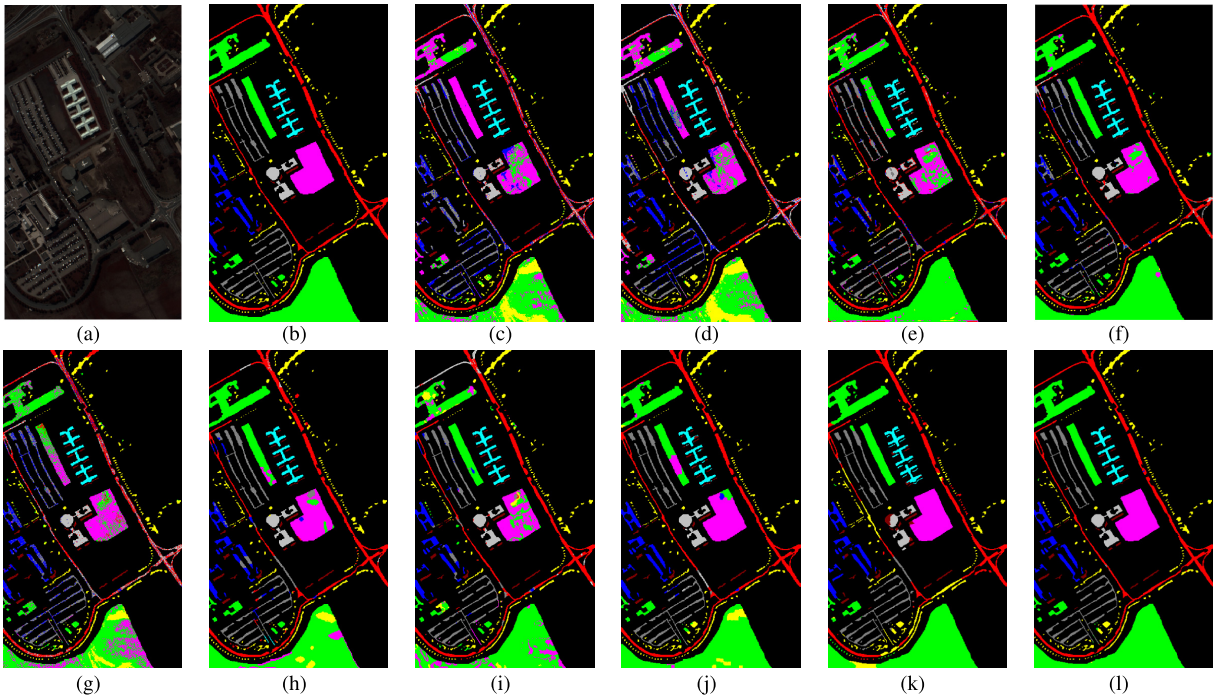


Fig. 9. Classification maps obtained by different methods on the Pavia University dataset. (a) False color image. (b) Ground-truth. (c) MLP. (d) SVM (RBF). (e) 3DCNN. (f) DRCNN. (g) GCN. (h) S^{2}GCN. (i) S^{2}GraphSAGE. (j) MDGCN. (k) DIGCN. (l) S^{3}FGCN.

TABLE IX

RESULTS OF DIFFERENT METHODS ON THE SALINAS DATASET

ID	MLP	SVM (RBF)	3DCNN	DRCNN	GCN	S^{2}GCN	S^{2}GraphSAGE	MDGCN	DIGCN	S^{3}FGCN
1	96.12±0.80	96.57±0.48	98.06±0.27	99.81±0.31	98.62±0.86	98.97±0.54	**100.00±0.00**	99.68±0.42	**100.00±0.00**	99.91±0.03
2	97.18±1.62	90.55±2.39	99.76±0.03	99.53±0.30	99.07±1.21	99.21±0.44	**99.83±0.29**	99.81±0.40	98.10±1.89	99.75±0.25
3	83.29±2.99	74.65±5.93	97.87±0.44	90.10±8.43	97.03±1.10	97.03±2.18	93.02±2.61	97.86±2.39	99.91±0.25	99.82±0.16
4	96.41±1.23	96.99±0.24	98.72±0.14	**99.94±0.13**	99.28±0.49	99.11±0.50	98.94±0.87	95.12±1.69	94.28±4.99	98.50±1.68
5	92.48±1.64	94.31±1.96	89.26±0.95	99.34±0.23	98.58±0.79	97.19±2.34	99.82±0.19	93.48±1.36	94.86±2.00	98.60±0.08
6	98.30±0.23	98.15±0.55	99.27±0.51	**100.00±0.00**	99.58±0.30	99.42±0.32	99.64±0.62	98.88±1.07	96.48±2.24	**100.00±0.00**
7	93.32±2.65	97.99±0.56	98.04±0.20	98.65±1.89	99.13±0.25	99.17±0.26	99.98±0.03	97.93±1.78	99.99±0.03	**100.00±0.00**
8	61.76±3.11	60.58±11.9	78.47±1.19	81.08±7.65	67.94±8.33	72.08±4.96	86.83±0.34	92.37±4.27	**97.65±1.81**	97.18±1.46
9	93.49±0.65	92.77±1.36	97.01±0.22	99.28±0.68	98.50±0.85	98.42±1.63	99.37±0.38	99.80±0.18	99.58±1.15	**99.83±0.13**
10	65.87±2.97	58.82±13.4	91.95±0.58	95.63±1.93	89.64±1.57	91.11±2.56	87.49±5.23	94.42±1.54	95.42±3.46	**98.02±1.96**
11	90.06±1.87	77.84±2.14	88.78±1.03	99.46±0.32	94.80±2.98	98.46±1.38	88.59±2.74	95.64±2.96	96.26±3.55	**99.66±1.07**
12	96.16±1.25	85.26±5.6	97.51±0.16	**100.00±0.00**	99.71±0.08	99.53±0.53	99.51±0.85	92.35±3.25	91.86±3.74	99.62±0.12
13	92.48±2.45	94.96±0.57	97.71±1.08	**100.00±0.00**	97.99±0.61	98.12±0.62	97.29±1.54	91.90±2.91	92.19±4.26	98.68±0.27
14	89.15±0.90	87.72±1.42	94.21±0.52	**99.96±0.05**	93.58±2.60	95.85±1.62	99.78±0.38	87.93±6.14	95.95±2.76	93.69±4.92
15	52.27±6.10	61.5±15.3	75.35±1.46	85.46±14.3	66.18±9.08	71.16±3.63	77.79±2.52	95.58±3.59	**98.44±1.58**	96.99±1.52
16	74.07±6.93	78.93±5.04	88.93±2.34	99.65±0.40	97.24±1.21	97.06±1.92	94.28±6.63	99.65±0.54	**100.00±0.00**	99.27±0.51
OA(%)	79.30±0.71	78.98±0.97	89.63±0.29	93.18±1.27	87.16±0.85	88.57±1.21	92.54±0.85	95.94±1.06	97.61±0.69	**98.54±0.16**
AA(%)	85.78±0.49	84.23±0.83	93.18±0.19	96.74±1.17	93.55±0.39	94.42±0.57	95.14±0.93	95.78±0.62	96.94±0.66	**98.72±0.27**
Kappa(%)	77.09±0.78	76.69±1.05	88.47±0.33	82.42±1.43	85.74±0.92	88.13±0.97	92.09±0.32	95.49±1.18	97.34±0.76	**98.38±0.18**

TABLE X

RESULTS OF DIFFERENT METHODS ON THE UNIVERSITY OF HOUSTON DATASET

ID	MLP	SVM (RBF)	3DCNN	DRCNN	GCN	S^{2}GCN	S^{2}GraphSAGE	MDGCN	DIGCN	S^{3}FGCN
1	84.01±0.99	83.67±0.72	93.70±0.38	95.62±5.41	88.16±1.90	**96.30±3.07**	72.31±2.06	93.42±4.25	93.07±2.73	92.33±2.75
2	18.58±5.80	83.40±8.10	97.52±0.39	96.78±3.92	97.20±0.48	**98.57±1.47**	75.12±0.92	93.67±3.60	94.17±2.93	93.55±3.31
3	90.45±2.52	93.69±0.23	97.65±0.39	96.75±1.83	97.91±0.13	98.09±1.74	53.99±0.55	98.12±1.09	95.00±1.68	**98.85±1.12**
4	92.72±1.11	93.05±1.15	94.96±0.38	93.41±3.23	96.55±0.41	**97.68±2.89**	63.34±1.82	95.58±1.85	90.47±4.09	97.09±0.93
5	82.69±3.7	87.31±3.22	96.13±0.38	99.15±0.78	89.79±0.71	97.66±1.12	83.41±1.54	99.00±1.30	**100.00±0.00**	97.65±1.48
6	66.77±0.49	66.27±1.26	68.72±0.28	93.83±2.22	98.21±1.15	**96.84±1.17**	82.05±1.24	93.28±6.08	94.10±3.86	94.92±0.86
7	45.47±5.71	31.51±6.22	88.19±0.35	80.71±6.26	73.67±1.94	83.48±5.89	23.00±1.92	87.68±4.41	**96.06±2.80**	82.88±3.89
8	41.09±2.36	35.88±1.84	77.81±0.31	78.30±5.43	65.71±4.64	76.15±4.37	56.38±1.78	80.45±6.12	73.36±5.63	**80.71±3.81**
9	71.21±3.18	75.02±1.87	79.79±0.32	76.91±5.62	70.27±3.03	82.17±1.78	62.14±4.86	89.64±2.26	**94.33±3.33**	92.30±1.35
10	24.04±14.60	2.44±1.42	85.76±0.35	81.99±7.04	74.71±2.32	86.85±8.32	71.45±1.11	90.06±6.41	88.76±7.63	**95.11±2.17**
11	57.99±8.03	69.64±1.01	84.53±0.34	84.04±4.86	75.36±2.37	88.57±5.06	62.73±2.23	86.73±3.22	**90.68±4.32**	89.12±3.65
12	11.21±9.03	15.67±14.00	86.18±0.34	81.92±8.31	79.29±4.80	78.64±4.79	67.64±4.04	**89.44±5.69**	87.08±4.25	83.72±3.04
13	11.43±15.10	12.62±13.90	58.77±0.24	86.54±2.58	12.09±2.68	75.62±6.93	63.33±4.32	92.78±4.45	**92.79±4.34**	84.38±6.55
14	88.51±0.69	87.38±0.93	94.39±0.38	99.31±1.18	86.03±3.31	99.45±0.44	42.13±4.47	99.43±0.97	**100.00±0.00**	**100.00±0.00**
15	84.88±2.66	91.94±0.20	99.09±0.40	**99.60±0.50**	95.29±1.67	98.03±1.07	60.60±3.13	96.27±1.72	97.90±1.62	98.22±0.63
OA(%)	61.12±1.59	60.62±1.47	88.20±0.35	88.08±1.09	80.35±0.61	89.31±1.00	62.86±0.27	91.40±0.92	91.72±0.64	**92.21±0.46**
AA(%)	52.07±1.40	61.97±1.78	86.90±0.35	89.66±0.95	80.02±0.46	90.33±1.06	62.67±0.20	92.37±0.89	92.52±0.16	**92.57±0.57**
Kappa(%)	58.19±1.68	57.63±1.63	87.23±0.35	87.10±1.18	78.72±0.66	88.44±1.08	62.86±0.27	90.70±1.00	91.03±0.69	**91.56±0.50**

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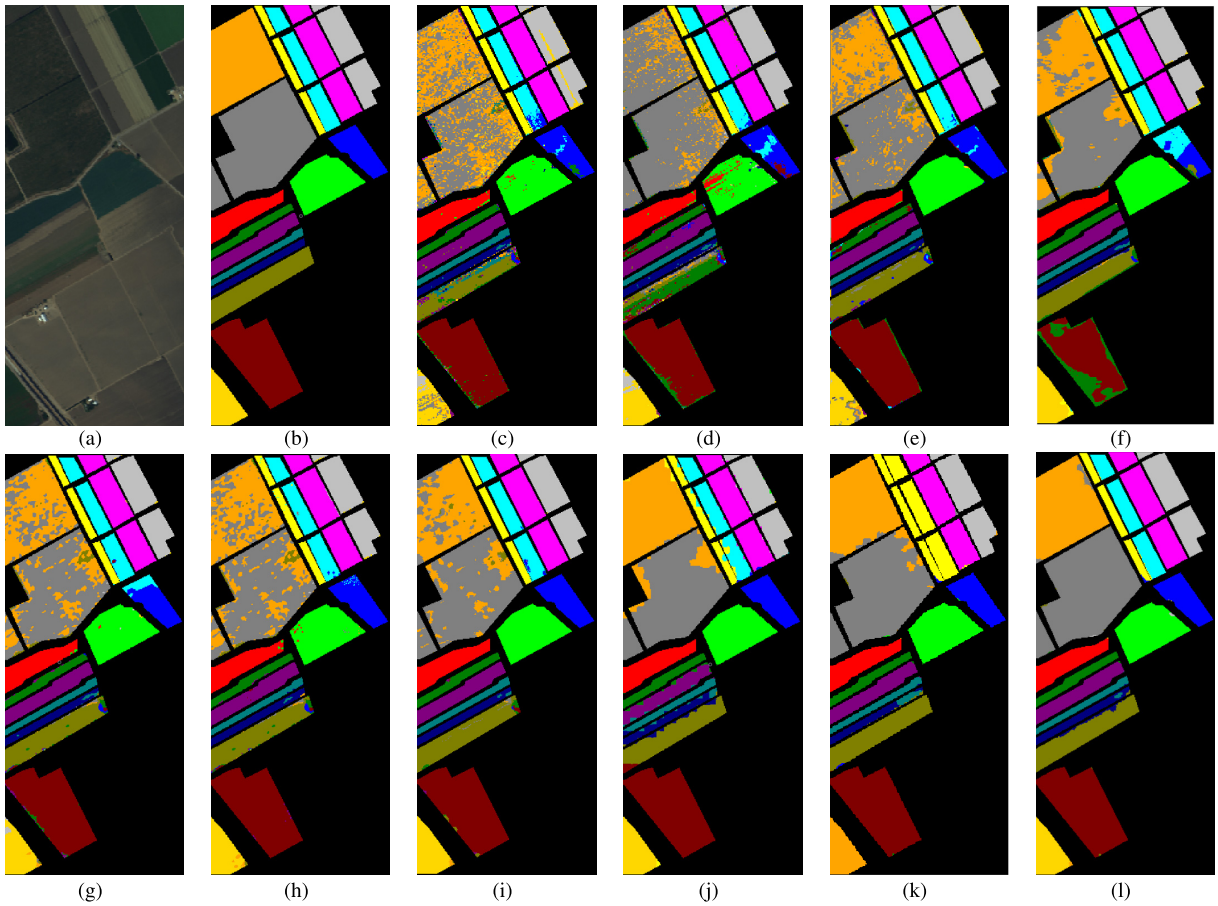


Fig. 10. Classification maps obtained by different methods on the Salinas dataset. (a) False color image. (b) Ground-truth. (c) MLP. (d) SVM (RBF). (e) 3DCNN. (f) DRCNN. (g) GCN. (h) S^2GCN. (i) S^2GraphSAGE. (j) MDGCN. (k) DIGCN. (l) S^3FGCN.

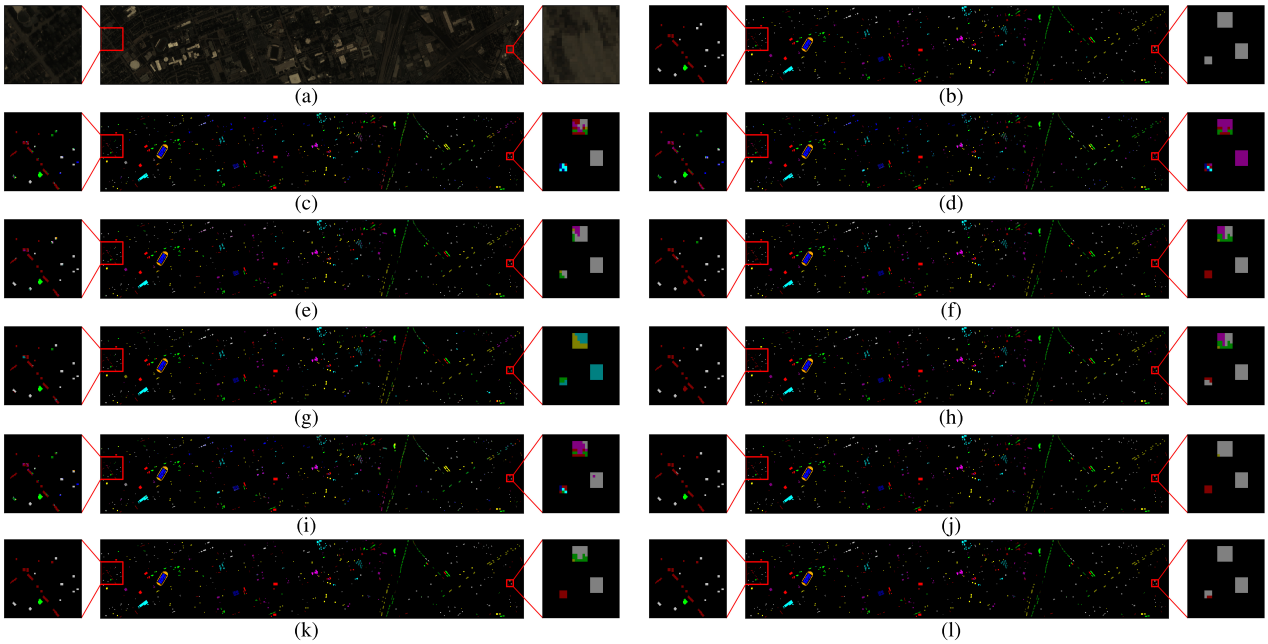


Fig. 11. Classification maps obtained by different methods on the University of Houston dataset. (a) False color image. (b) Ground-truth. (c) MLP. (d) SVM (RBF). (e) 3DCNN. (f) DRCNN. (g) GCN. (h) S^2GCN. (i) S^2GraphSAGE. (j) MDGCN. (k) DIGCN. (l) S^3FGCN.

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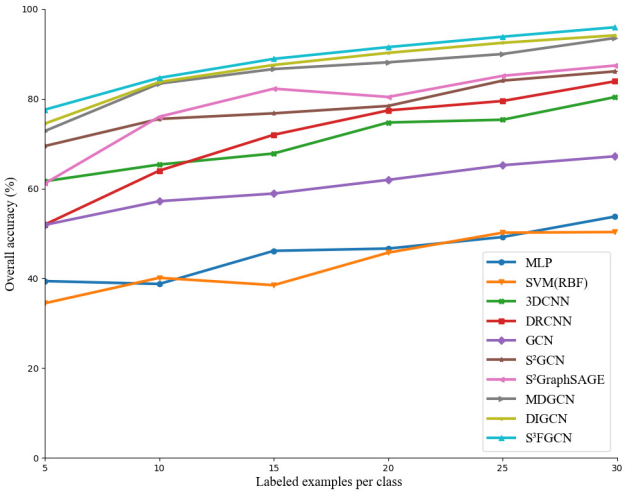


Fig. 12. Overall accuracies of various methods on the Indian Pines dataset under different numbers of labeled examples per class.

TABLE XI

OA (%), AA (%), AND KAPPA COEFFICIENT (%) UNDER DIFFERENT MODEL SETTINGS ON FOUR DATASETS

OA				
MODULES	IP	PaviaU	Salinas	Houston
$Spectral-GCN$	74.50	88.58	96.16	77.98
$Spatial-GCN$	76.40	88.87	95.97	77.18
S^2-GCN	92.71	97.46	96.85	88.48
$Spectral-GCN+Spatial-GCN$	92.51	97.10	97.22	88.21
$Spectral-GCN+S^2-GCN$	93.27	89.15	97.14	87.78
$Spatial-GCN+S^2-GCN$	92.97	97.50	96.65	90.71
S^3FGCN *Without Dynamic Sampling*	94.16	97.91	98.15	90.66
S^3FGCN	**95.97**	**98.53**	**98.54**	**92.21**
AA				
MODULES	IP	PaviaU	Salinas	Houston
$Spectral-GCN$	73.90	88.58	96.67	79.91
$Spatial-GCN$	71.21	84.87	96.44	78.43
S^2-GCN	95.71	98.25	98.08	88.78
$Spectral-GCN+Spatial-GCN$	92.51	98.12	98.29	88.57
$Spectral-GCN+S^2-GCN$	96.17	84.98	97.31	88.77
$Spatial-GCN+S^2-GCN$	96.12	98.41	97.91	90.88
S^3FGCN *Without Dynamic Sampling*	96.81	98.50	98.77	91.64
S^3FGCN	**97.59**	**99.13**	**98.72**	**92.57**
Kappa				
MODULES	IP	PaviaU	Salinas	Houston
$Spectral-GCN$	70.94	88.58	95.73	76.18
$Spatial-GCN$	72.92	85.41	95.17	75.31
S^2-GCN	91.65	96.65	96.50	87.53
$Spectral-GCN+Spatial-GCN$	91.80	96.21	96.91	87.24
$Spectral-GCN+S^2-GCN$	92.30	85.74	97.14	86.77
$Spatial-GCN+S^2-GCN$	91.96	96.70	96.82	89.94
S^3FGCN *Without Dynamic Sampling*	93.31	97.24	97.94	89.90
S^3FGCN	**95.38**	**98.05**	**98.38**	**91.56**

(S^2-GCN) with dynamic sampling, respectively. To demonstrate the contribution of each module, we compared the performance of one module, two modules, and all the three modules with and without dynamic sampling, leading to eight combinations. The results are shown in Table XI.

We see that in the case of single module, the most important one is the spectral–spatial graph convolution module. When we combine the modules and apply the three graph convolutions together, the model accuracy improves significantly. After applying the dynamic sampling idea to the model, the performance of AA improves by 1.81% on the Indian Pines dataset, by 1.62% on the Pavia University dataset, and by an average of 0.97% on the other two datasets. Then, to illustrate the effectiveness of dynamic sampling, we calculated the overall accuracies of each 500 epochs with and without dynamic sampling on four datasets. From Fig. 13, we can see that without dynamic sampling, the model converges earlier to a lower accuracy. In the early stage of training, the initial training accuracies of two settings are similar, and the performance may even be slightly better without dynamic sampling on some datasets. The reason is that there are many misclassified samples in early stage of training, including relatively easy to classify samples. With the training going on, dynamic sampling pays more attention to samples that are hard to classify, so that the model can continue to focus on less discriminative samples while learning good feature representations and leading to better performance.

TABLE XII

RESULTS OF DIFFERENT SUPERPIXEL SEGMENTATION METHODS ON THE INDIAN PINES DATASET

ID	MDGCN	DIGCN	S^3FGCN WITH SLIC	S^3FGCN WITH HMS
1	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
2	85.68±3.51	90.92±4.27	91.90±2.33	**92.25±1.94**
3	84.08±2.85	94.87±3.92	94.94±2.82	**96.19±2.51**
4	98.84±2.14	98.32±1.93	**99.52±1.23**	99.13±1.61
5	93.51±4.14	95.29±3.44	96.29±1.91	**97.07±2.22**
6	**97.23±2.10**	95.95±1.85	95.20±4.70	97.16±1.26
7	89.5±9.26	96.36±4.03	**100.00±0.00**	**100.00±0.00**
8	99.82±0.38	99.84±0.37	**100.00±0.00**	**100.00±0.00**
9	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
10	90.15±4.49	89.58±4.95	**93.39±2.90**	92.88±2.68
11	93.39±2.99	91.75±3.52	93.73±1.78	**95.38±1.42**
12	92.57±2.56	**93.81±3.73**	93.20±2.14	92.95±3.38
13	99.43±1.21	99.65±0.68	**98.92±1.84**	98.52±2.00
14	97.39±3.47	98.94±1.73	**100.00±0.00**	**100.00±0.00**
15	99.02±0.60	98.58±1.60	**100.00±0.00**	**100.00±0.00**
16	97.14±2.09	99.48±0.72	**100.00±0.00**	**100.00±0.00**
OA(%)	93.56±0.91	94.16±1.07	95.31±0.66	**95.97±0.63**
AA(%)	95.36±0.50	96.41±0.54	96.95±1.52	**97.59±0.41**
Kappa(%)	92.64±1.03	93.34±1.21	94.63±0.75	**95.38±0.72**

To verify the effectiveness of the HMS method, we compared it with SLIC on the Indian Pines dataset in the experiments. The experimental results are shown in Table XII, which indicates that the HMS method is better than the SLIC superpixel segmentation method in OA, AA and Kappa. Overall, S^3FGCN can extract common information under different embeddings. In addition, the experiments on dynamic sampling prove that paying attention to the sample points that are hard to classify has a very important impact on the training of the whole model.

G. Running Time

Table XIII shows the running time of the traditional models and deep learning models on four datasets. All the codes were written based on python. Each model framework was implemented on PyTorch, and the experiments were run under the Linux system. The resources used for the experiments included a computer with a 3.40-GHz Intel Core i7 central processing unit (CPU), 32-GB RAM, and a GTX 1080 Ti graphics processing unit (GPU). It can be seen from the table that the running time of the CNN-based model is generally longer because it requires deep network structure. The traditional models have great advantages in speed, but their accuracy is relatively low. The GCN-based model network uses a fixed graph to reduce the network parameters and the network depth is shallow, so the training time is short. However, as the number of nodes increases in the data preprocessing stage, the required memory and the time to construct the adjacency matrix also increase greatly. The speed of our model

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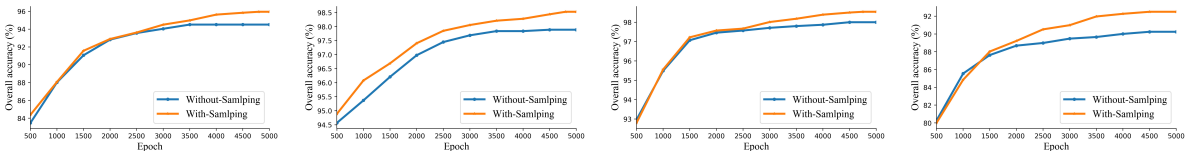


Fig. 13. Overall accuracies on four datasets with and without dynamic sampling. (a) Indian Pines dataset. (b) Pavia University dataset. (c) Salinas dataset. (d) University of Houston dataset.

TABLE XIII
RUNNING TIME COMPARISON (IN SECONDS) OF DIFFERENT METHODS

Dataset	MLP	SVM (RBF)	3DCNN	DRCNN	GCN	S^2GCN	S^2GraphSAGE	MDGCN	DIGCN	S^3FGCN
IP	24	0.17	448	408	21	34	4572	27	53	195
PaviaU	17	0.25	1565	638	33	54	7636	93	187	264
Salinas	25	0.82	2559	910	62	93	12255	36	616	254
Houston	18	1.06	213	443	32	41	3968	34	124	456

is slightly higher than MDGCN because MDGCN only uses spectral features in the spatial neighborhood. Compared with DIGCN, our method takes slightly higher time on most of the datasets. On the Salinas dataset, the class distribution is even. There are fewer superpixels, so our running time is lower than DIGCN. We extracted the features separately from the spectral and spatial domains, extracted the common features in the joint spectral–spatial domain, and finally perform feature aggregation. The steps have stronger information expression capability and lead to higher accuracy than other compared methods. In summary, our model can maintain its speed well while ensuring accuracy.

V. CONCLUSION

In this paper, we have introduced a superpixel spectral–spatial feature fusion graph convolution network for hyperspectral image classification (S^3FGCN). Our method not only learns different information from the spatial and spectral domains separately but also learns common information in the joint spectral–spatial domain. Moreover, the proposed method also learns the spatial information under different spatial neighborhoods with more extensive implicit information. In addition, the use of superpixels greatly reduces the problem of excessive memory caused by the construction of the adjacency matrix. Our method focuses on points that are difficult to classify through dynamic sampling, which is beneficial to model learning. Extensive experiments show that our proposed method is better than several state-of-the-art methods. In the future, we will explore more efficient deep learning methods to reduce the training time.

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